Yorgos Pantis

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I am a 3rd-year Ph.D. student at the University of Athens, co–affiliated with the Archimedes Unit, where I work on theory-driven machine learning under the supervision of Christos Tzamos. I am also undertaking research visits to MIT CSAIL. My research focuses on neural learning and reasoning across architectures, including neural networks and transformers.

Education

Ph.D. in Machine Learning Nov. 2024 – Curr.

University of Athens, Dept. of Informatics and Telecommunications

- Research interests: Reasoning in LLMs and Neural Networks theory
- Supervisor: Christos Tzamos
- Publications: **NeurIPS** (1), **ICLR** (1); 2 additional papers under review at **ICLR**.
- Total 9-month research visit at MIT

M.Sc. in Statistical Science Oct. 2020 – Sep. 2021

University of Oxford, Dept. of Statistics, Lincoln College

- Thesis: *Authorship Attribution in Ancient Greek Texts*
- Supervisors: Yannis Assael, Yorgos Deligiannidis, and Thea Sommerschield
- Grade: **Merit**

B.Sc. in Mathematics Sep. 2016 – Aug. 2020

University of Patras, Dept. of Mathematics

- Thesis: *Mathematical Techniques in Data Science and Machine Learning* – Grade: **10.00/10.00**
- Supervisor: Michael N. Vrahatis
- Grade: **8.90/10.00** (Distinction), equivalent to first-class honors (**top 1%**)

Experience

Ph.D. Researcher Jul. 2024 – Curr.

Athena Research Center, Archimedes Unit

- Conducting theory-based research on reasoning in Transformer models and Neural Networks theory
- Supervisor: Christos Tzamos

Visiting Researcher Feb. 2027 – May 2027

MIT, Computer Science and Artificial Intelligence Laboratory Mar. 2026 – May 2026

- Investigating reliability and robustness in generative models
- Supervisor: Constantinos Daskalakis

Research Assistant Apr. 2024 – Jun. 2024

Technical University of Denmark, Dept. of Applied Mathematics and Computer Science

- Explored Deep Learning theory via Riemannian geometry
- Supervisor: Georgios Arvanitidis

Research Assistant Apr. 2023 – Dec. 2023

Czech Technical University in Prague, Faculty of Electrical Engineering

- Focused on the Mini-batch Stochastic Subgradient Projection method
- Supervisors: Vyacheslav Kungurtsev, Jakub Mareček

Research Assistant Apr. 2022 – Mar. 2024

Athena Research Center, Institute for the Management of Information Systems

- Researched the interface of NLP, Privacy, and Security tasks using LLMs
- Supervisor: Constantinos Patsakis

Research Assistant Oct. 2021 – Mar. 2022

Max Planck Institute for Mathematics in the Sciences

- Worked on authorship attribution focusing on Plutarch's texts using LLMs
- Supervisors: Ivan P. Yamshchikov, Jürgen Jost

Publications

C: Conference J: Journal †: equal contribution

- [C_5] *Beyond Labels: PAC Learning with Intermediate Supervision*
  Nikos Tsikouras, Yorgos Pantis, Constantinos Daskalakis, Christos Tzamos. **ICLR 2027 (Under Review)**
- [C_4] *SGD Selects Initialization-Dependent Representation in Overparameterized ReLU Networks*
  Nikos Tsikouras, Yorgos Pantis, Ahmed Mehdi Inane, Mehrab Hamidi, Ioannis Mitliagkas. **ICLR 2027 (Under Review)**
- [C_3] *A Derandomization Framework for Structure Discovery: Applications in Neural Networks and Beyond*
  Nikos Tsikouras, Yorgos Pantis, Ioannis Mitliagkas, Christos Tzamos. **ICLR 2026**
- [$C_2^\dagger$] *Teaching Transformers to Solve Combinatorial Problems through Efficient Trial & Error*
  Panagiotis Giannoulis, Yorgos Pantis, Christos Tzamos. **NeurIPS 2025**
- [J_1] *The Anatomy of Deception: Measuring Technical and Human Factors of a Large-scale Phishing Campaign*
  Anargyros Chrysanthou, Yorgos Pantis, Constantinos Patsakis. **Computers & Security 2024**
- [C_1] *BERT in Plutarch's Shadows*
  Ivan P. Yamshchikov, Alexey Tikhonov, Yorgos Pantis, Charlotte Schubert, Jürgen Jost. **EMNLP 2022**

Honors & Awards

Full 4-Year Ph.D. Scholarship Jul. 2024 – Curr.
Athena Research Center, Archimedes Unit
- Competitive doctoral fellowship awarded based on academic excellence and research potential

Visiting Research Grant Jan. 2027 – Jun. 2027
Bodossaki Foundation
- Funded research visit in the US, collaborating with MIT and Harvard researchers

Gold Reviewer Award May 2026
International Conference on Machine Learning
- Recognized for exceptional peer-review quality and service; included complimentary conference registration

Research Infrastructure Grant Dec. 2024 – May 2026
Greek Research & Technology Network
- Granted access to national high-performance computing infrastructure to support doctoral research

Full Master's Scholarship Oct. 2020 – Aug. 2021
Bodossaki Foundation
- Awarded for outstanding undergraduate academic performance to pursue a highly competitive master's program

COVID-19 Assistance Fund Jun. 2021
University of Oxford, Lincoln College
- Received financial support through a competitive student assistance program during the pandemic

Vivian Green Student Assistance Scholarship Feb. 2021
University of Oxford, Lincoln College
- Awarded in recognition of academic progress during master's studies

Full 4-Year Undergraduate Scholarship Sep. 2016 – Sep. 2020
Greek State Scholarships Foundation
- Scholarship awarded annually on the basis of academic excellence throughout undergraduate studies

Consistently Ranked in the Top 1% of Cohort Sep. 2016 – Aug. 2020
University of Patras, Dept. of Mathematics
- Maintained a top 1% class ranking throughout the entire bachelor's program

Ranked 2nd Among Admitted Students Sep. 2016
University of Patras, Dept. of Mathematics
- Placed 2nd among all students admitted to the Mathematics undergraduate program

Talks & Service

Talks

Transformer Reasoning

- Greek Stochastics π' Conference (Oral Presentation) — *Jul. 2026*
- Harvard University — *May 2026*
- Northeastern University — *May 2026*
- Boston University — *Apr. 2026*
- Massachusetts Institute of Technology — *Apr. 2026*
- HSBC — *Sep. 2025*
- Athena Research Center, Archimedes Unit — *May 2025*

AI & Education Outreach

- Greeks in AI Conference — *Jul. 2026*
- Greek Institute of Education Policy — *Jan. 2026*

Reviewing

- NeurIPS, ICLR, ICML, TMLR, AAAI — *2024 – Curr.*

Teaching Assistant

University of Athens, Dept. of Informatics and Telecommunications

- Computational Learning Theory — *Fall 2025*
- Artificial Intelligence II (Deep Learning for NLP) — *Fall 2025*

Volunteer

Athena Research Center, Archimedes Unit — *Sep. 2025 – Curr.*

- Committee member for Greece's National AI Competition; preparing students for European and International Olympiads